

More on t - Part 2

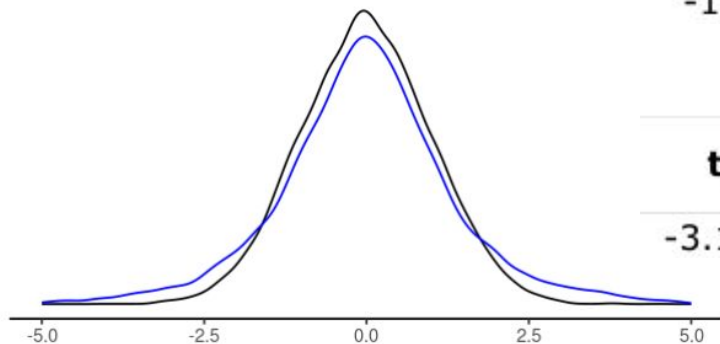
MATH 213

Group 1

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4

t = blue, z = black



z_q025

<dbl>

-1.954854

z_q25

<dbl>

-0.679019

z_q75

<dbl>

0.6671864

z_q925

<dbl>

1.449163

t_q025

<dbl>

-3.167306

t_q25

<dbl>

-0.7927017

t_q75

<dbl>

0.7564919

t_q925

<dbl>

1.949499

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

The middle 50% and 95% of the t-statistics cover a wider range than the middle 50% and 95% of z-statistics

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3.2

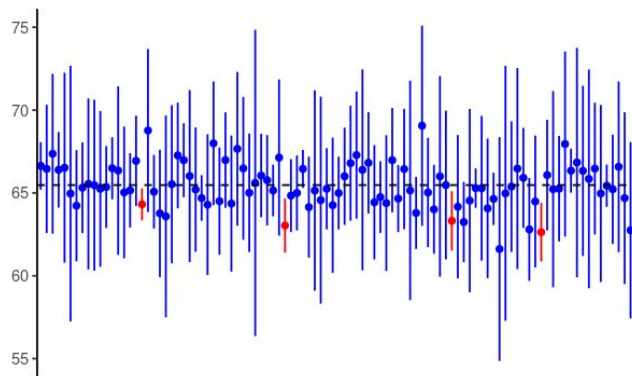
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+/- 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446 - very

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?



96

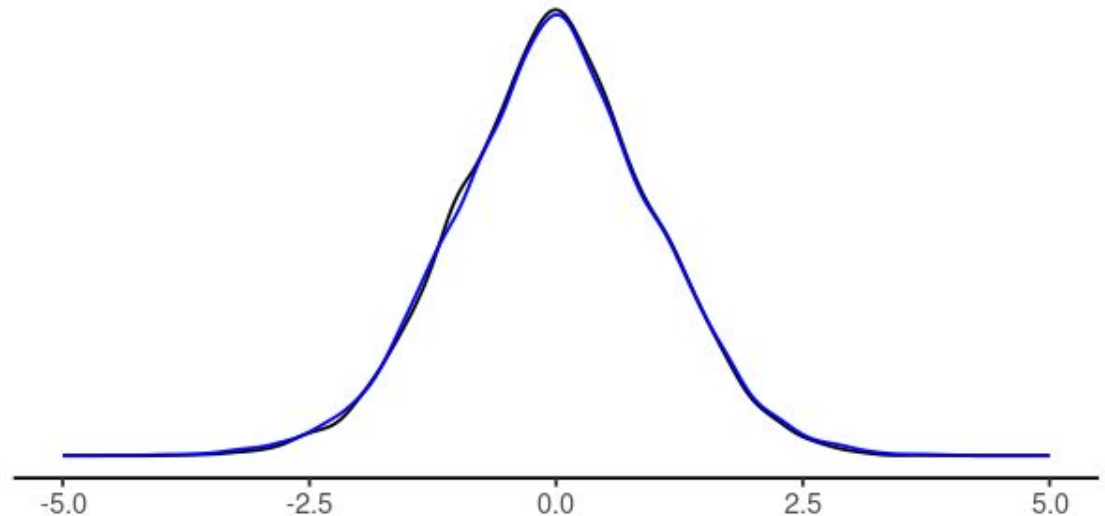
Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

No

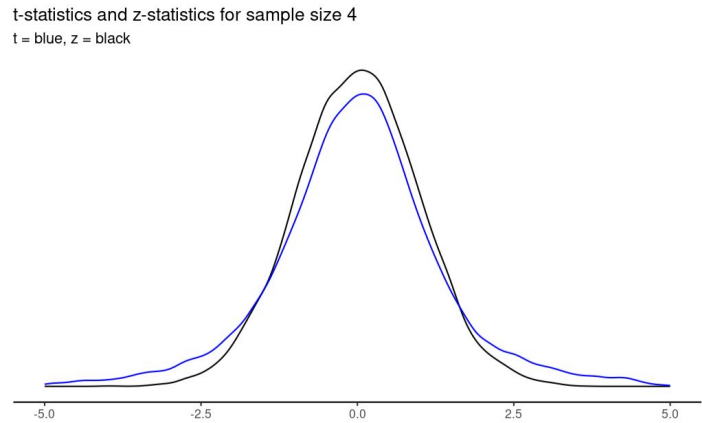
t-statistics and z-statistics for sample size 30

t = blue, z = black



Group 2

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.



A tibble: 1 × 4

Z_q025 <dbl>	Z_q25 <dbl>	Z_q75 <dbl>	Z_q975 <dbl>
-1.918504	-0.6548642	0.6666331	1.957124

1 row

A tibble: 1 × 4

T_q025 <dbl>	T_q25 <dbl>	T_q75 <dbl>	T_q975 <dbl>
-3.152047	-0.7465931	0.749435	3.219506

1 row

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

On the graph, the t-statistic has a less extreme mode which results if a less concentrated data (compared to the z-statistic) leading to a larger spread and larger number. This is reflected in the quantiles, even though Z_q025 & Z_q25 are larger than their T counter parts, the T quantiles reflect the larger range.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

~3

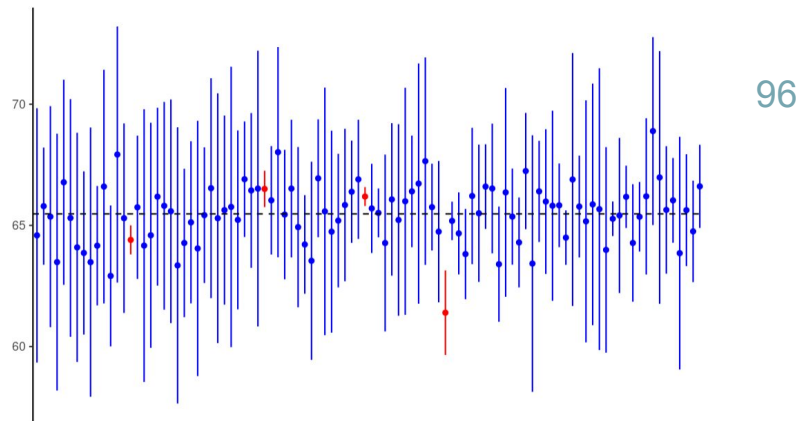
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+/- 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446 - pretty close

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?

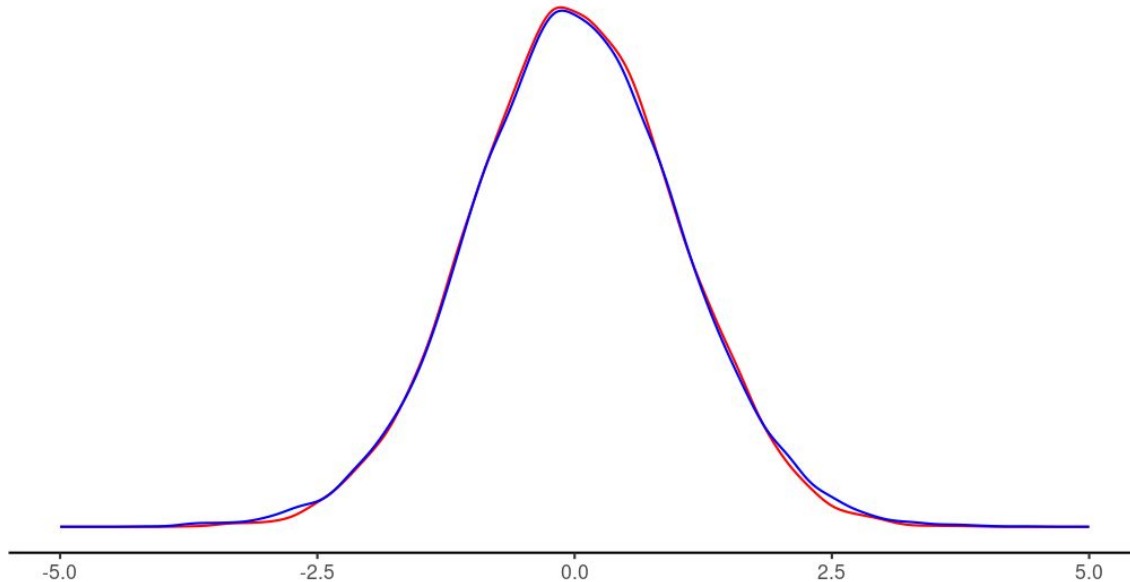


Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

t-statistics and z-statistics for sample size 30

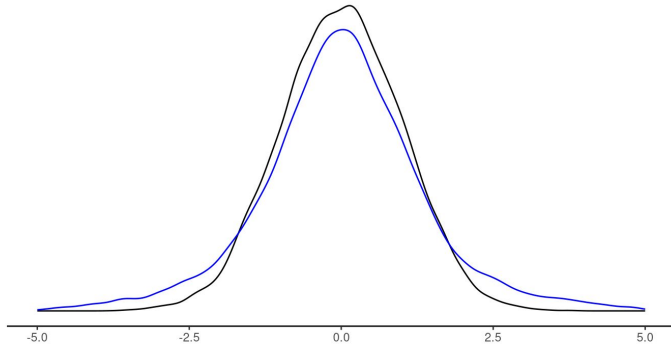
t = blue, z = red



Group 3

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



Z-
Q1: -0.674749
Q3: 0.682306
Z_low95: -1.922285
z_high95: 1.921857

T-
Q1: -0.7624964
Q3: 0.795474
T_low95: -3.194873
T_high95: 3.256239

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

In the graph the t-statistic is flatter and wider than the z curve, especially in the tails of each. The 95% interval for the t-statistic is much wider compared to the z-statistic, show extreme t-values occur more.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3.2 to capture 95%

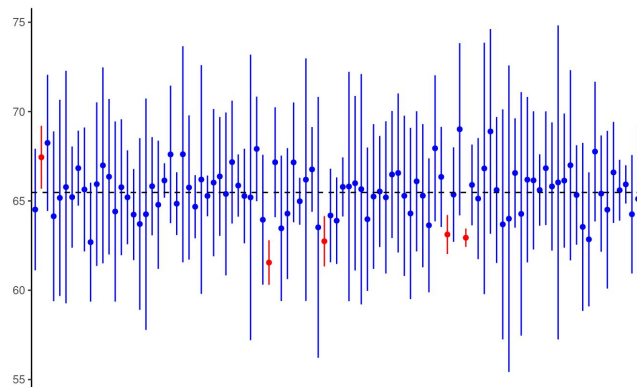
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than ± 2 approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?

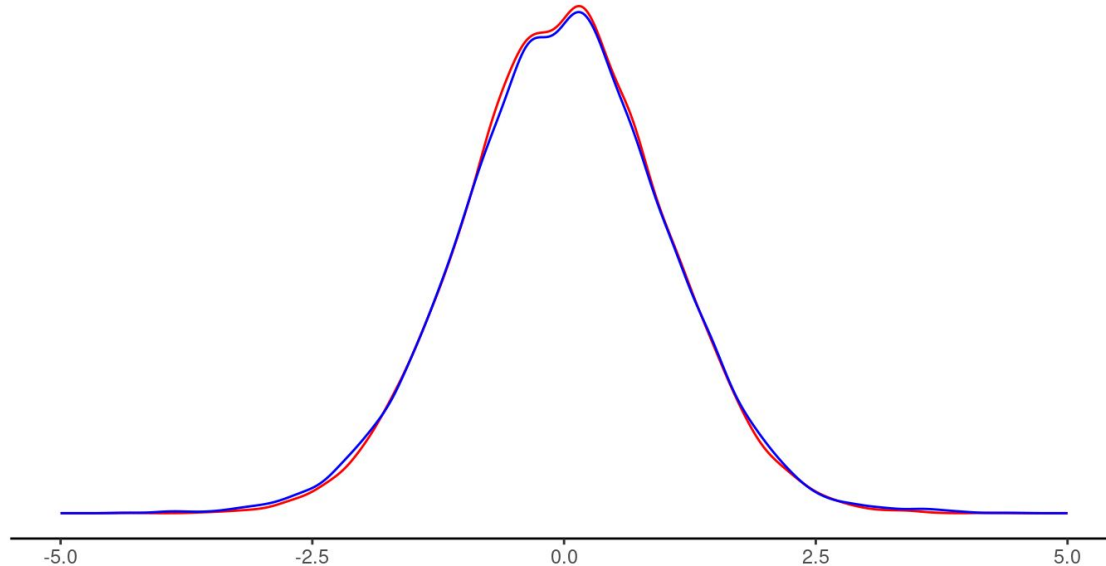


Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

t-statistics and z-statistics for sample size 30

t = blue, z = red



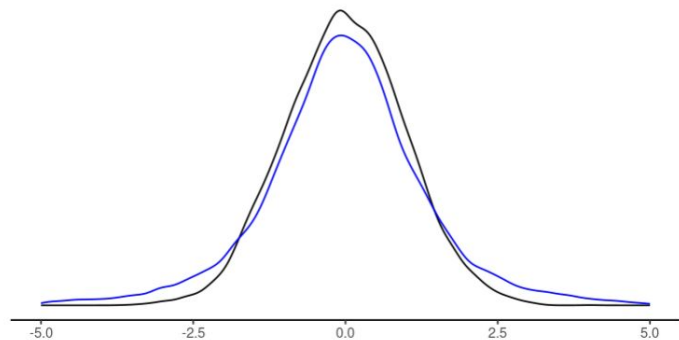
The notable differences I see is that the z stat is steeper while the t stat is wider.

Group 4

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4

t = blue, z = black



A tibble: 1 × 5				
z_q025 <dbl>	z_q25 <dbl>	z_mean <dbl>	z_q75 <dbl>	z_q975 <dbl>
-1.944131	-0.6777603	0.006636411	0.6744864	1.986125
1 row				

A tibble: 1 × 5				
t_q025 <dbl>	t_q25 <dbl>	t_mean <dbl>	t_q75 <dbl>	t_q975 <dbl>
-3.111008	-0.7760165	0.01022377	0.7476055	3.254621
1 row				

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

Large t-statistics are more likely than large z-statistics because almost all t-statistics are between -3.1 and 3.25, while almost all z-statistics are between -1.9 and 1.9.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3.25?

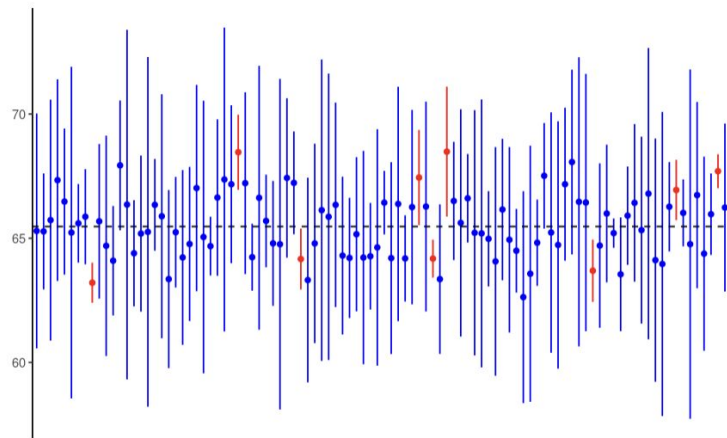
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To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than ± 2 approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?



91/100

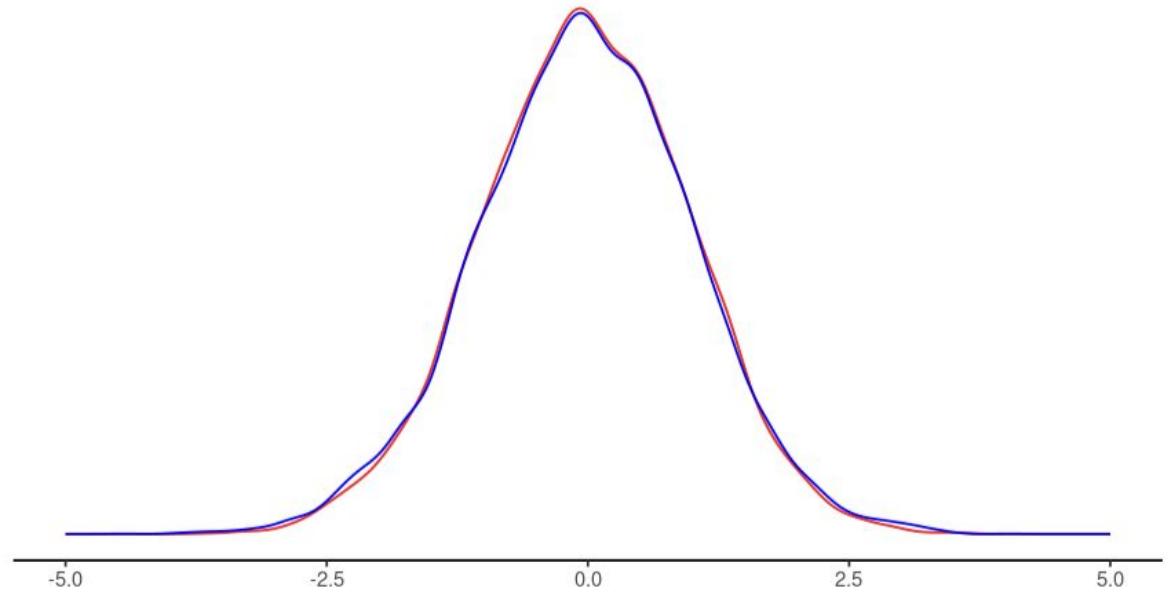
Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

There are no notable differences between the t-scores and z-scores for this sample size of 30.

t-statistics and z-statistics for sample size 30

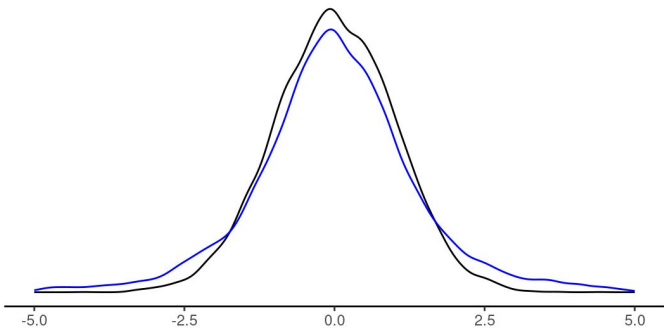
t = blue, z = red



Group 5

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



zq025 <dbl>	zq25 <dbl>	zq75 <dbl>	zq975 <dbl>
-1.98495	-0.6751232	0.6909198	1.923618

tq025 <dbl>	tq25 <dbl>	tq75 <dbl>	tq975 <dbl>
-3.141456	-0.7404972	0.7857356	3.374766

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?
- Well in the graph, the blue line seems to be a bit less of a high curve when in comparison to the black. And on the side of the quantiles, the sizes in comparison to the z quantiles for each one turns out to be $t > z$ in pretty much all cases by whatever margin.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

4.295

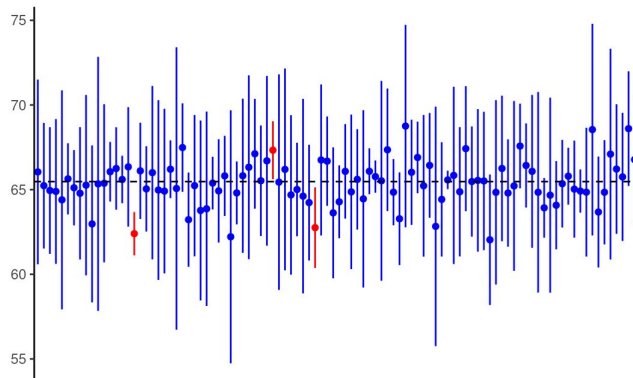
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+/- 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446, surprisingly closer than i thought it would be.

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter? 97

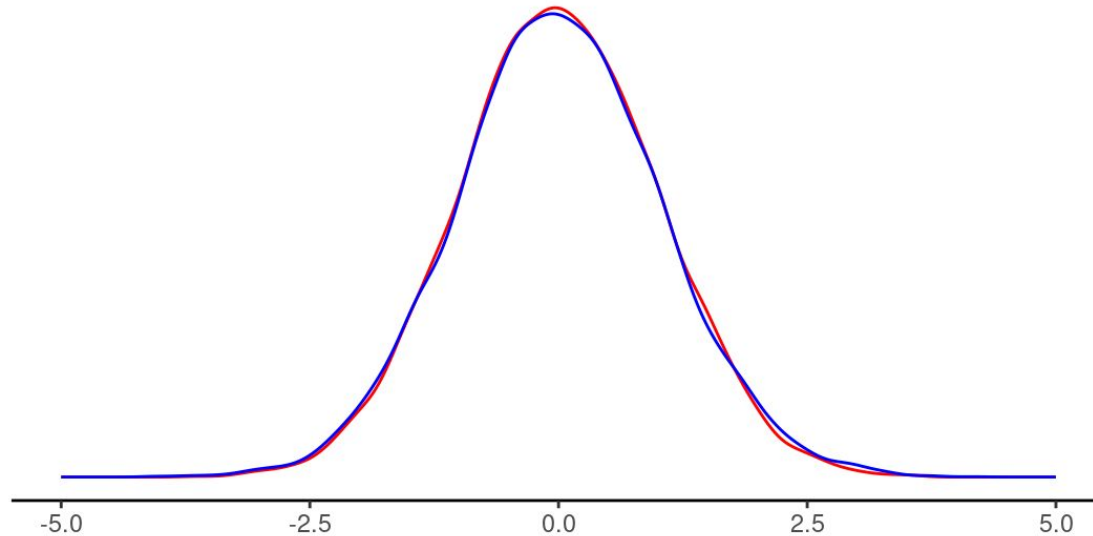


Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions? nope

t-statistics and z-statistics for sample size 30

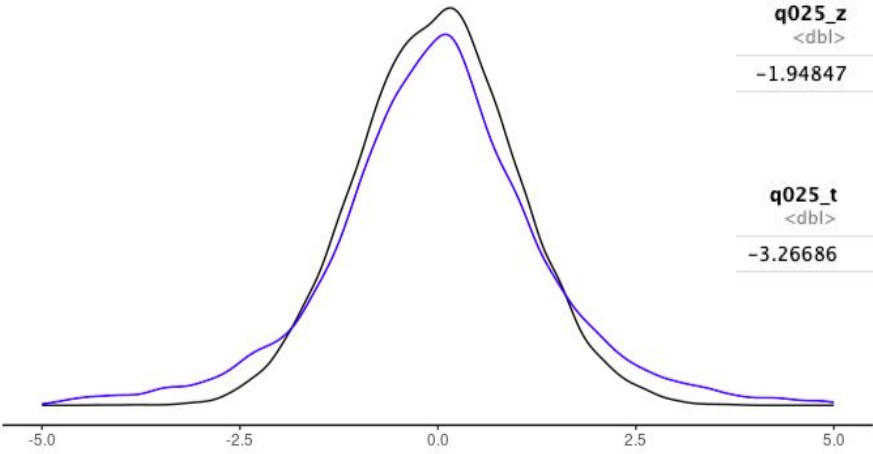
t = blue, z = red



Group 6

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



q025_z <dbl>	q25_z <dbl>	q75_z <dbl>	q975_z <dbl>
-1.94847	-0.6884459	0.6644959	1.982259

q025_t <dbl>	q25_t <dbl>	q75_t <dbl>	q975_t <dbl>
-3.26686	-0.7764531	0.7725804	3.172154

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?
The quartile values for the t-statistic are larger and the graph demonstrates a curve that is smaller than the z-statistic.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3, since the interval would span from -3 to +3 and include the 0.025 and 0.975 quantile values.

Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+ / - 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command ``qt(p = 0.975, df = 3)``. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.18

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?

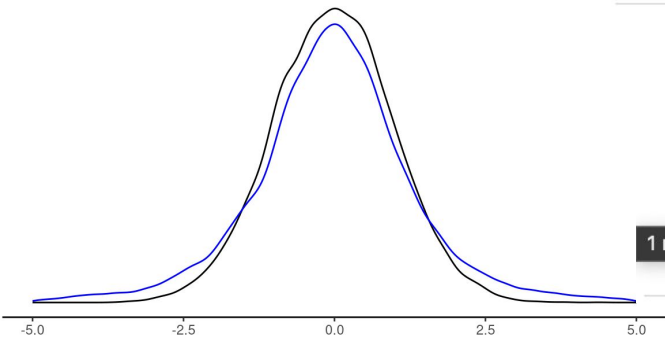
Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

Group 7

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



Z	q025 <dbl>	q25 <dbl>	q75 <dbl>	q975 <dbl>
	-1.98246	-0.7022656	0.6419647	1.913648

t	q025 <dbl>	q25 <dbl>	q75 <dbl>	q975 <dbl>
1 row	-3.305368	-0.7771112	0.7347154	3.160043

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

T-statistics are more likely to be larger because their range ranges from -3.3 - 3.2, while the z-statistic rangers from -1.9 - 1.9

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3.16

Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

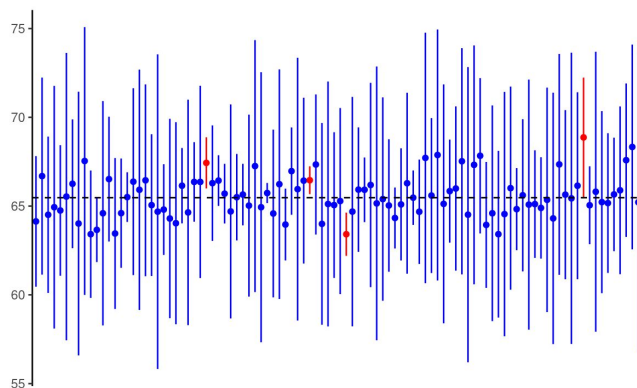
To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than ± 2 approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?

96

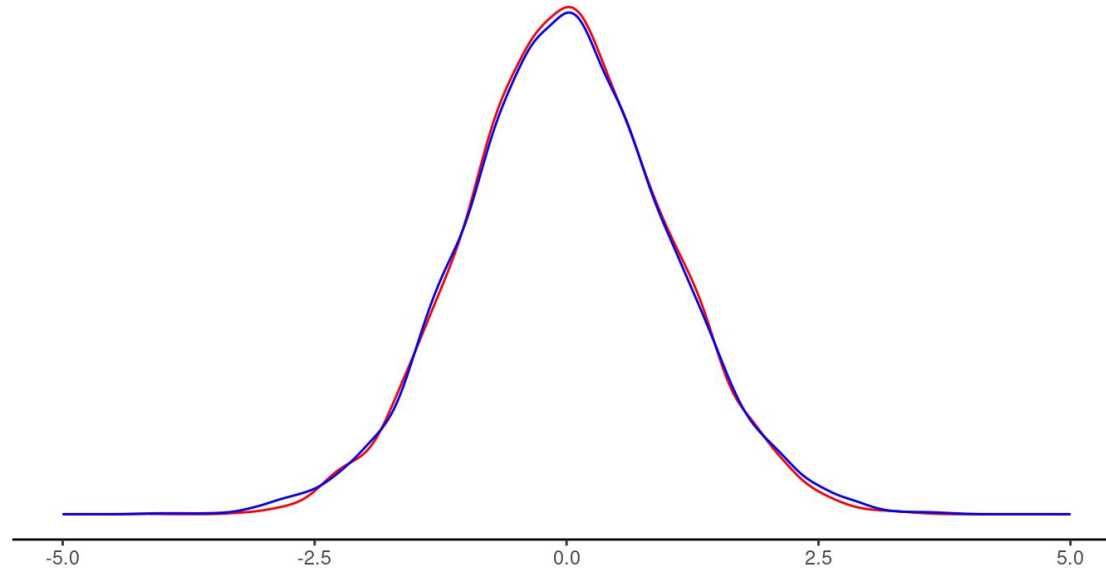


Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

Not much difference

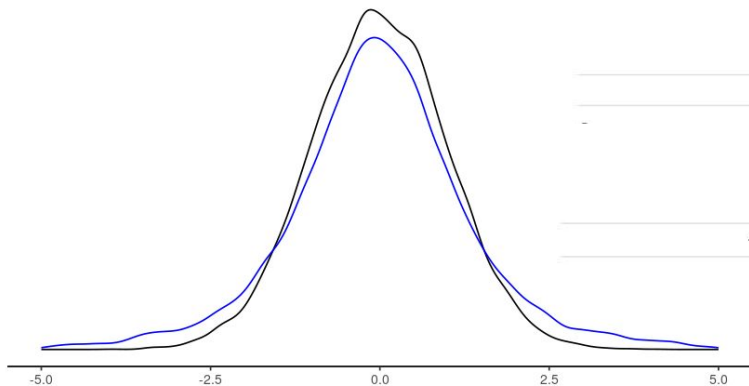
t-statistics and z-statistics for sample size 30
t = blue, z = red



Group 8

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



Q1_z <dbl>	Q3_z <dbl>	q025_z <dbl>	q975_z <dbl>
-0.6787277	0.6581627	-1.958268	1.92159
Q1_t <dbl>	Q3_t <dbl>	q025_t <dbl>	q975_t <dbl>
-0.7704734	0.7572069	-3.248304	3.246806

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

Comparing the Q1 and Q3 values for each statistic respectively (and same with the 0.025 and 0.975 quantiles), the t-statistic tends to have a larger value. The graph also shows that there are more t-statistic points at larger values (negative or positive).

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

3.2

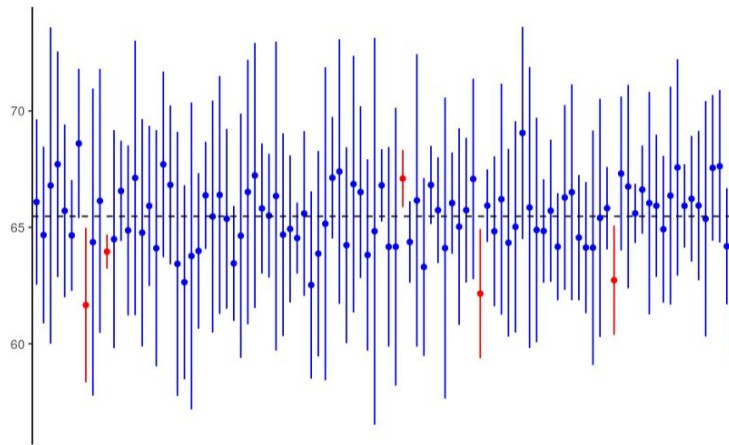
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+/- 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

3.182446. Very close.

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter? **95**

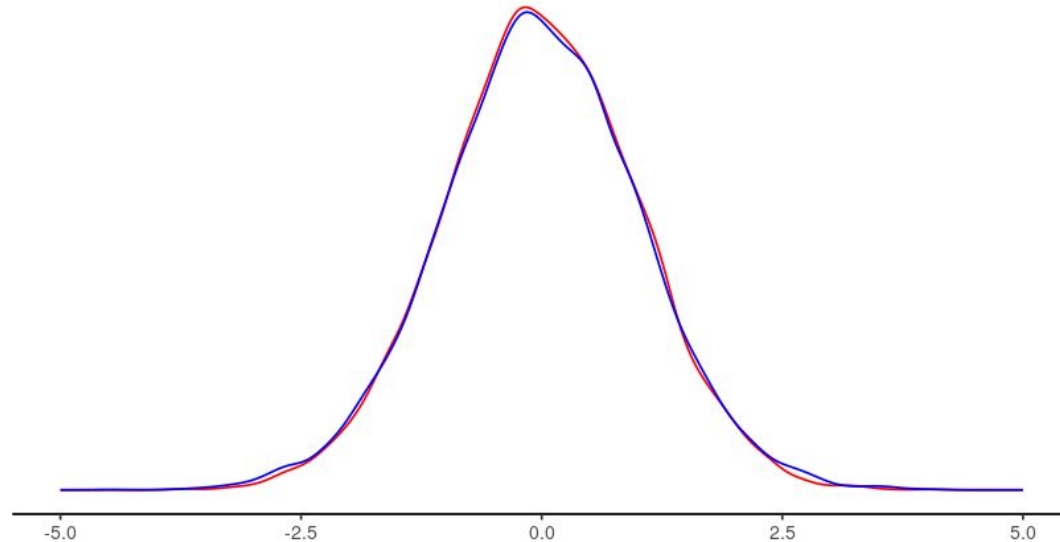


Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions? **Not very noticeable differences.**

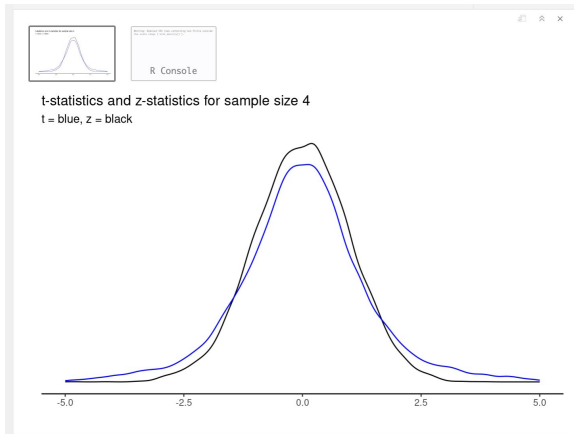
t-statistics and z-statistics for sample size 30

t = blue, z = red



Group 9

- Put your density plot comparing etc. for these two distributions h



```
67 q025 = quantile(t, 0.025),
68 q25 = quantile(t, 0.25),
69 q75 = quantile(t, 0.75),
70 q95 = quantile(t, 0.975)
71
72 ```
```

A tibble: 1 × 5

mean <dbl>	q025 <dbl>	q25 <dbl>	q75 <dbl>	q95 <dbl>
0.01573825	-3.1789	-0.7669103	0.7643559	3.292929

1 row

```
73
74 ```{r}
75 samples_4 %>%
76   summarize(
77     mean = mean(z),
78     q025 = quantile(z, 0.025),
79     q25 = quantile(z, 0.25),
80     q75 = quantile(z, 0.75),
81     q95 = quantile(z, 0.975))
82 ```
```

A tibble: 1 × 5

mean <dbl>	q025 <dbl>	q25 <dbl>	q75 <dbl>	q95 <dbl>
-0.003133087	-1.995231	-0.6728399	0.6689488	1.919172

1 row

o put your quartiles

T score on top
z on bottom

- What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?
After reviewing the quantile ranges form th

For any given quantile range between T and Z, we can see the value of the bound for T is larger than Z.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+ / - 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command ``qt(p = 0.975, df = 3)``. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made?

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?

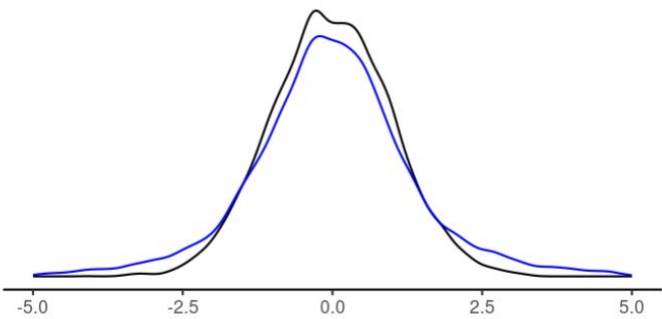
Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

Group 10

1. Put your density plot comparing the z- and t-distributions for samples of size 4 here. Also put your quartiles etc. for these two distributions here.

t-statistics and z-statistics for sample size 4
t = blue, z = black



T:

A tibble: 1 × 4			
q025 <dbl>	q25 <dbl>	q75 <dbl>	q975 <dbl>
-3.239973	-0.795453	0.760392	3.241352
1 row			

Z:

A tibble: 1 × 4			
q025 <dbl>	q25 <dbl>	q75 <dbl>	q975 <dbl>
-1.937987	-0.6885457	0.6701268	1.956571
1 row			

2. What about the graphs and quantiles shows that large t-statistics are more likely than large z-statistics?

We can see that the t-statistics are larger because if you take a look at the quantiles and if you take any one of them and compare it to the z-statistics quantile the t will end up being larger. For example, the data underneath 75% for the t is .76 while the z is .67 which is smaller.

In our first work with confidence intervals, we found that using 2 standard errors would give us confidence intervals that captured the population parameter for about 95% of samples. Looking at your work on t-statistics, you can see why 2 approximate standard errors does not work: the interval from -2 to +2 does not include the entire middle 95% of t-statistics.

3. Judging from your 0.025 and 0.975 quantiles, how many approximate standard errors would you recommend that we use for 95% confidence intervals based on the standard error estimated from the sample standard deviation?

We should approximate 4 standard errors

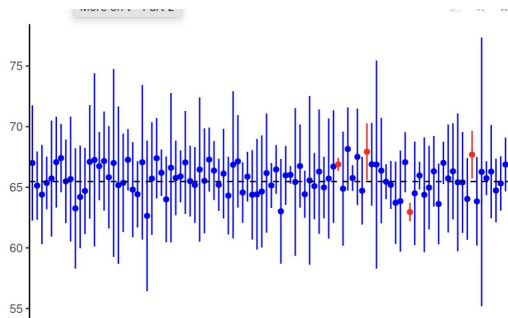
Gosset found that for *any* normally distributed population, t-statistics of samples of size 4 follow the same distribution that you see here. This distribution is called the "t-distribution with 3 degrees of freedom."

To make confidence intervals for small samples, we should follow Gosset's advice. Looking at his "t distribution", we see that we need something more than $+/- 2$ approximate standard errors to capture almost all (95% of) samples. You made a suggestion above for how many approximate standard errors we should use.

4. To find the official number according to Gosset's mathematics, use the R command `qt(p = 0.975, df = 3)`. The letter "q" stands for "quantile", and "df" stands for "degrees of freedom". Paste your output here. How close is it to the suggestion you made? My suggestion was off by .8 with my suggestion of 4 standard errors

```
## {r}
qt(p = 0.975, df = 3)
##
[1] 3.182446
```

5. Put here your graph showing 100 confidence intervals for samples of size 4, calculated using Gosset's t-procedure. How many of your 100 intervals included the population parameter?



Finally, we compare the z- and t-distributions using a larger sample size. We *should* see that for larger sample sizes, the z- and t-distributions are comparable. We do not have the same problems as we had before.

6. Put your density plot comparing the z- and t-distributions for samples of size 30 here. Do you see any notable differences between the two distributions?

The notable differences between the 2 graphs is that This graph with bigger sample sizes ends up looking The same and only having minor differences. The Previous graph showed that the t-statistic didn't have A peak as big as the z-statistic but the t had larger values

t-statistics and z-statistics for sample size 30
t = blue, z = red

